# 19-200-0105B/19-200-0205A
# Basic Mechanical Engineering

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# Syllabus

**19-200-0105B/19-200-0205A BASIC MECHANICAL ENGINEERING**

**Course Outcomes:**

On completion of this course the student will be able to:

1. Understand basics of thermodynamics and working of steam turbines
2. Understand basics of internal combustion engines, refrigeration and air conditioning
3. Gain knowledge on the working of hydraulic turbines and centrifugal pumps
4. Identify manufacturing methods encountered in engineering practice and understand mechanism of power transmission

**Module I**

**Thermodynamics**: Thermodynamics systems – open, closed and isolated systems, equilibrium state of a system, property and state, process, cycle, Zeroth law of thermodynamics- concept of temperature, temperature scales. First law – internal energy, enthalpy, work and heat, Different processes (isobaric, isochoric, isothermal, adiabatic and polytropic processes). Second law – Kelvin-plank and Claussius statements and their equivalence, Carnot Cycle (Elementary problems only). Thermodynamic properties of Steam, Steam Generator. Different types of boilers, boiler mountings and accessories. Formation of steam at constant pressure, working of steam turbines, compounding of turbines.

**Module II**

**Internal Combustion Engines**: Air standard cycles – Otto and Diesel cycles, working of two stroke and four stroke Petrol and Diesel engines, Carburetted and MPFI engines, fuel pump, fuel injector, ignition system, cooling system, lubricating system.

**Refrigeration & Air-conditioning**: Introduction to refrigeration and air -conditioning, Rating of refrigeration machines, Coefficient of performance, Simple refrigeration vapour compression cycle (Elementary problems only), Summer and winter air conditioning.

**Module III**

**Hydraulic Turbines & Pumps**: Introduction, Classification, Construction details and working of Pelton Wheel, Francis and Kaplan turbines, Specific speed (Definition and significance only),

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Classification of water pumps, working of centrifugal pumps and reciprocating pumps (Theory of working principles only)

**Power plants:** Hydro-electric power plants, Thermal power plants, Nuclear power plants, Diesel power plants, Wind mills, solar energy (Working principles using schematic representations only)

**Module IV**

**Introduction to Manufacturing Systems**: Welding- different types of welding, resistance welding, arc welding, gas welding, Brazing and soldering, Different welding defects. Casting- different casting processes, sand casting, casting defects, Rolling- hot rolling and cold rolling, two high, three high , cluster rolling mills, wire drawing, forging, extrusion, Heat treatment of steel, elementary ideas of annealing, hardening, normalizing, surface hardening.

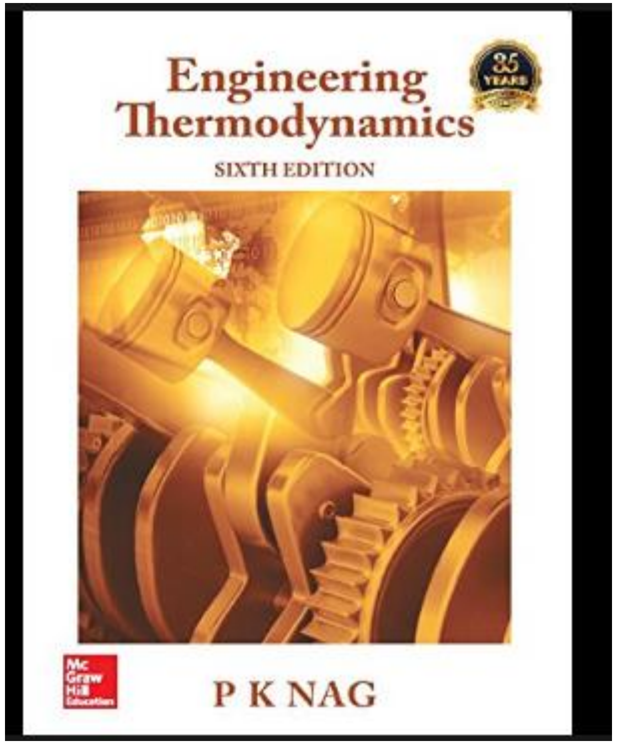
**Power Transmission Methods and Devices**: Introduction to Power transmission, Belt, Rope, Chain and Gear drive. Length of belt open and crossed. Ratio of belt tensions (Elementary problems only). Different types of gears (Elementary ideas only). Types and functioning of clutches.

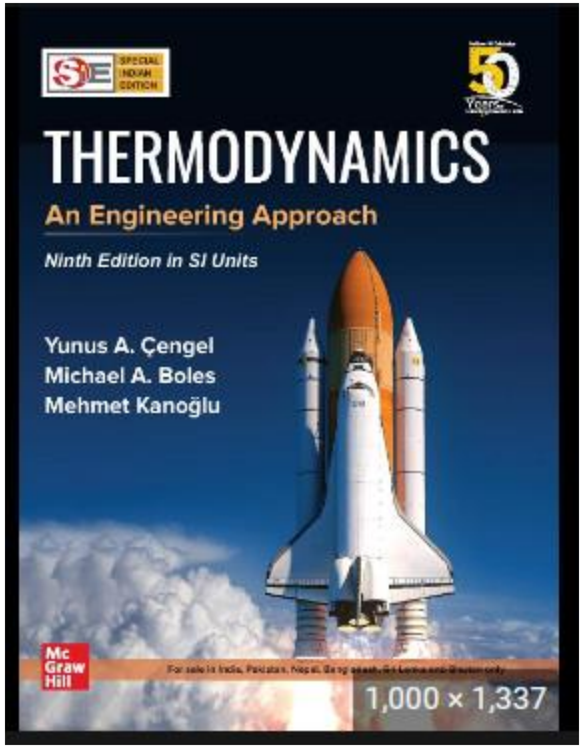
**References**


1. Nag, P.K. Engineering thermodynamics. (Fifth Edition). McGraw Hill Education (India) Pvt. Ltd, New Delhi.(2013).
2. Gill, J.H. Smith Jr. and Ziurys, E.J. Fundamentals of internal combustion engines, Oxford & IBH, New Delhi.(1959)
3. Stoecker, W. F. Refrigeration and air conditioning. Tata McGraw Hill, New Delhi. (1980).
4. JagadishLal. Hydraulic machines. Metropolitan Book co, New Delhi.(1994)
5. Raghavan, V. Material science and engineering, Prentice Hall of India, New Delhi. (2004)
6. Rajendar Singh.Introduction to basic manufacturing processes and workshop technology, New Age International, New Delhi. (2006).

# Module-1: Thermodynamics

# Text Books for Module I




**Note: Always refer standard textbooks mentioned in the syllabus for detailed reading**

# Thermodynamics

- Science of energy transfer and its effect on the physical properties of substances
- It is based upon observations of common experience which have been formulated into thermodynamic laws
- Thermodynamics:- Science of energy
- Therme:- Heat
- Dynamics:- Force
- One of them is ‘Law of conservation of energy’

# Example: Food Vs Exercise

# Example: Food Vs Exercise

- Input (Food) < Output (Exercise):- Weight loss

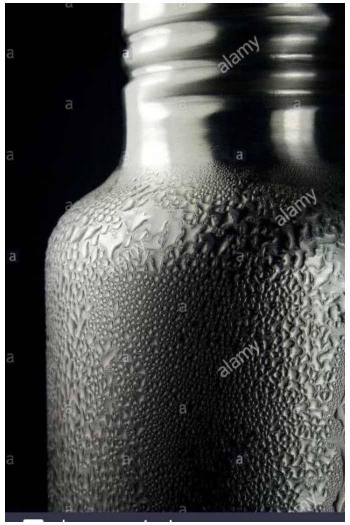
- Input (Food) > Output (Exercise):- Weight gain

# Thermodynamics:- Applications


# Thermodynamics:- Applications

Engines

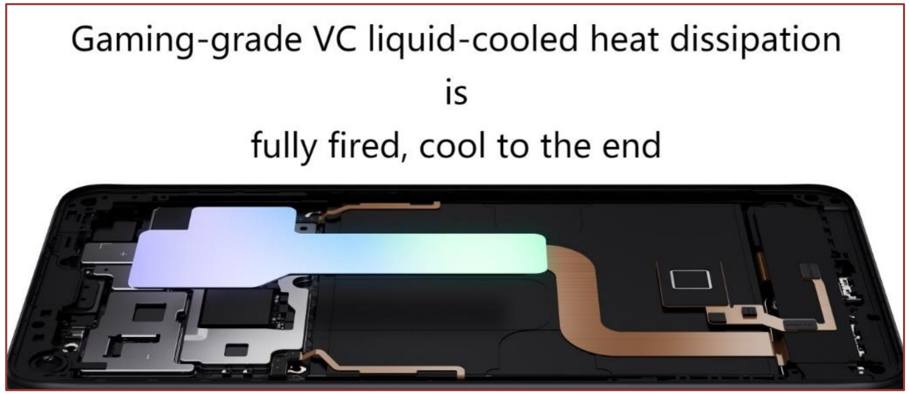
Radiator in cars



Condensation

Boiling

Electronic cooling

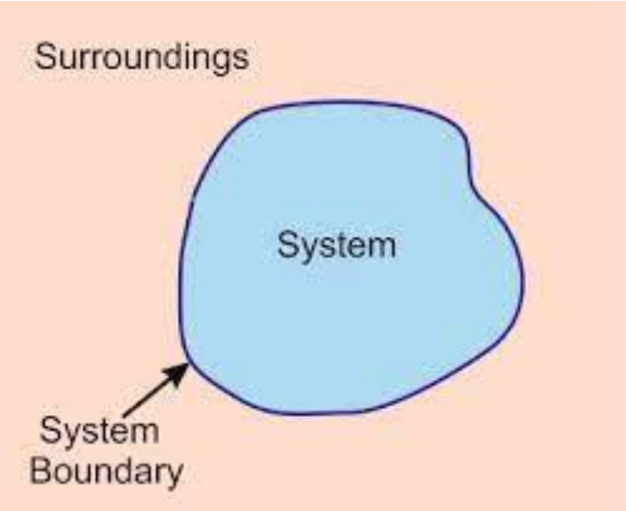



Heat Pipe in Realme phone

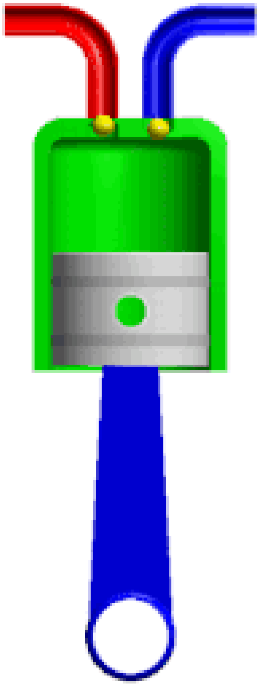
Cooling tower in power plant

# Classification of Systems

- System:-Quantity of matter/ a region of space chosen for study
- Surrounding:- Mass or region outside the system
- Boundary:- Real/imaginary surface that separate system from its surrounding, can be fixed or movable



# Moving/Flexible boundary?



# Energy interactions

- Heat:- Form of energy that is transferred between two systems (system with surrounding) by virtue of temperature difference

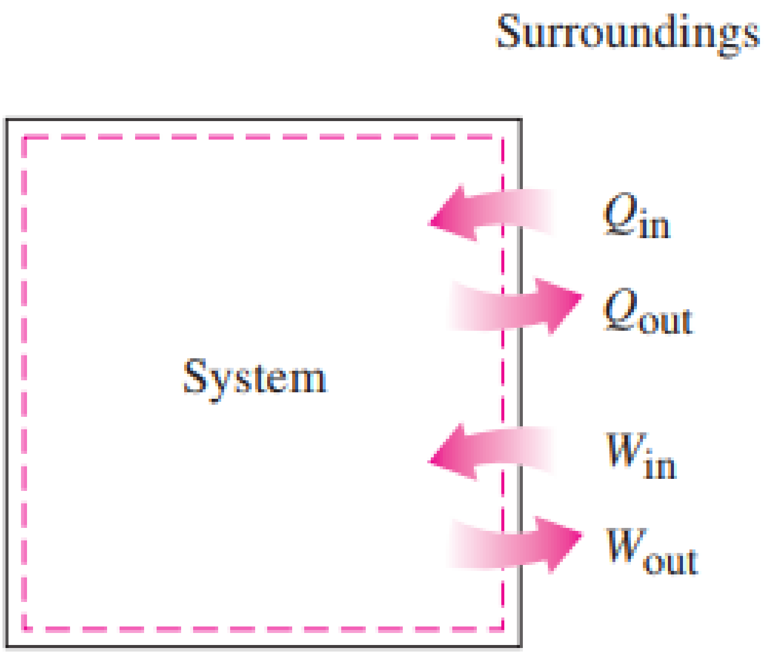
- Heat is energy in transition, it is recognized only as it crosses the boundary of a system

- Adiabatic system:- No heat transfer
  - Either system is well insulated
  - Or System and Surrounding are at same temperature

# Energy interaction

- If energy crossing the boundary of a closed system is not heat, then it must be work
- Work:- Energy transfer associated with a force acting through a distance
- Eg: Rising piston, rotating shaft etc.

# Energy interaction



# Type of system?


# Dimensions and Units

- Any physical quantity can be characterized by dimensions.
- The magnitudes assigned to the dimensions are called units.
- Primary/Fundamental dimensions
- Derived dimensions

The seven fundamental (or primary) dimensions and their units in SI

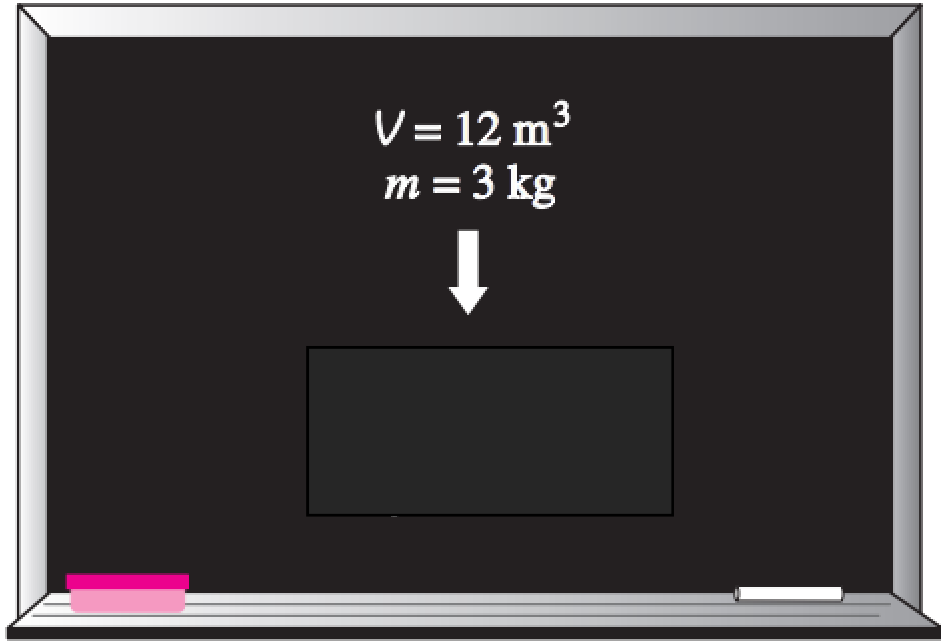
| Dimension | Unit |
|---|---|
| Length | meter (m) |
| Mass | kilogram (kg) |
| Time | second (s) |
| Temperature | kelvin (K) |
| Electric current | ampere (A) |
| Amount of light | candela (cd) |
| Amount of matter | mole (mol) |

# Derived Quantities

- Velocity
- Acceleration
- Force
- Work
- Pressure

# Few Properties

- Density $\rho = \dfrac{m}{V} \quad (\mathrm{kg/m^3})$
- Specific volume $v = \dfrac{V}{m} = \dfrac{1}{\rho}$
- Specific gravity $\mathrm{SG} = \dfrac{\rho}{\rho_{H_2O}}$
- Specific weight $\gamma_s = \rho g$



# Problem

For a SG = 13.6; Calculate

- Density
- Specific volume
- Specific weight

$$\rho = \frac{m}{V} \qquad (\mathrm{kg/m^3})$$

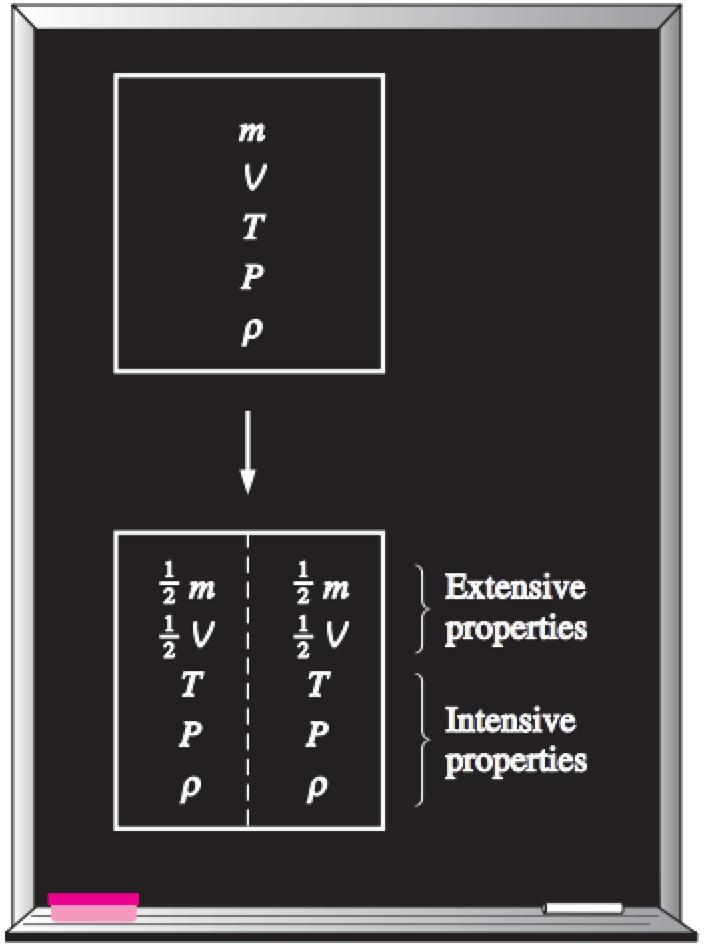
$$v = \frac{V}{m} = \frac{1}{\rho}$$

$$\gamma_s = \rho g$$

# Properties of a system

- Characteristics of a system is known as Properties
  - Ex: P, T, V etc.
- Extensive properties:- Value depends on size/extent of size
- Intensive properties:- Independent of size
- How to identify extensive and intensive?

# Properties of system



# Approaches in Thermodynamics

- Substance:- large number of particles

- Macroscopic approach / Classical thermodynamics
  - No need to know information of individual particles
  - Direct and easy way

- Microscopic approach / Statistical thermodynamics
  - More elaborated approach based on the average behaviour of individual particles

# Approaches in Thermodynamics

- Example:

- Pressure of a gas:- momentum transfer between particles and walls

- Individual particle behaviour is not required to know the pressure of the system

# Continuum

- Matter:- made of atoms, that are widely spaced
- It is convenient to disregard the atomic nature of a substance and view it as a continuous, homogeneous system with no holes;- that is Continuum
- This approach allow us to treat matter as point functions
- This approach is valid as long as the size of the system we deal is large relative to space between particles
- At very high vacuum the mean free path may be very large (eg: 0.1m), at that cases continuum approach is not valid, rarified gas theory should be used

# State and Equilibrium

- When a system not undergoing any change, all properties can be measured and it gives us a set of properties that describes condition or state of the system
- State:- Property values are fixed
- If one of them change; state will change
- Equilibrium:- state of balance, no unbalanced potential with in the system
- Thermodynamics deals with Equilibrium states
- A system in Equilibrium experience no changes when it is isolated from its surroundings

# Equilibrium

- Equilibrium
  - Thermal:- Temperature does not change
  - Mechanical:- No unbalanced force
  - Chemical:- Chemical composition does not change
  - Phase:- Mass of each phase does not change

# Thermodynamic Process

- If any one or more properties of the system undergo a change due to energy or mass transfer we say that the system has undergone a change of state
- The successive change of state of the system due to energy or mass transfer defined by definite path is called a process.
- The curve joining the successive state represents the process path
- If a system undergoes two or more processes and returns to its original state after conclusion of processes, the system is said to have undergone a cycle
- Process – change from one equilibrium state to another

# Point Function Vs Path Function

**Point Function:**

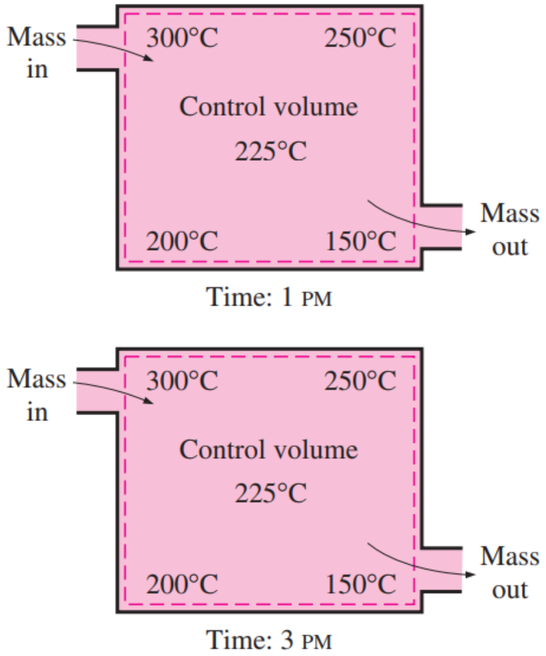
- When two co-ordinates are located on the graph, They define apoint and the two properties on the graph define state. These properties (p, T, v) are called point function.
- They depend on the state only, and not on how a system reaches that state.

**Path Function:**

- If the value of thermodynamic variable depends upon the path followed in going from one state to another, then the variable is a path function

# Steady Flow Process

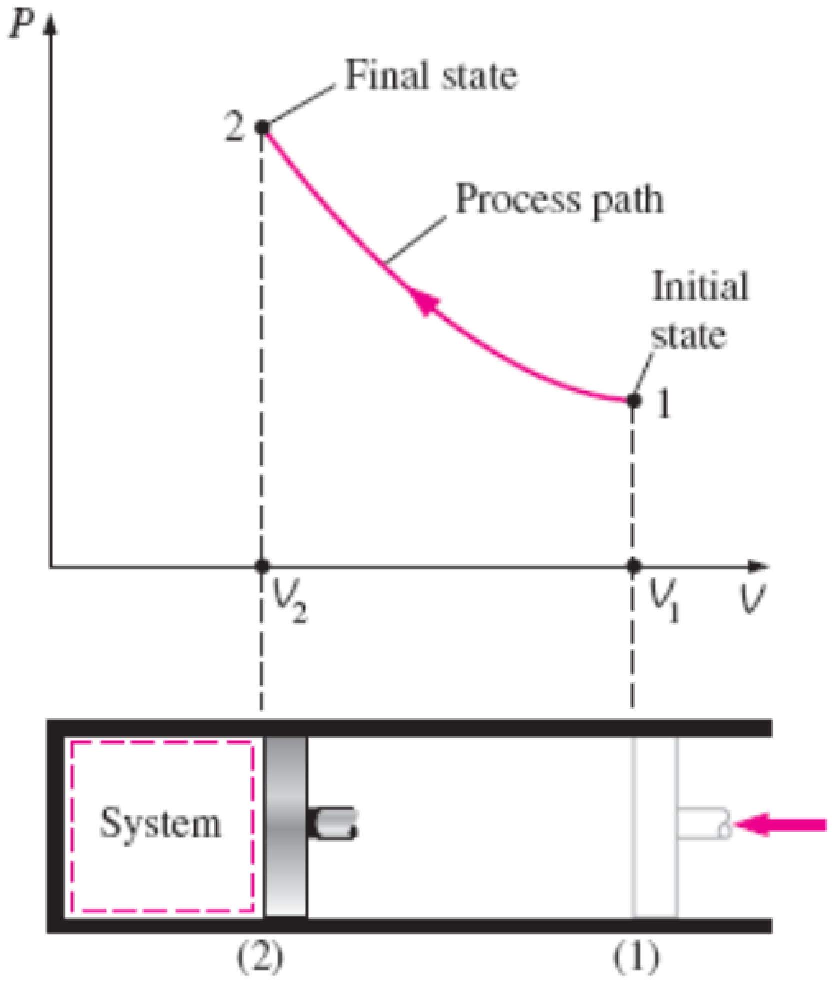
- Steady:- No change w.r.t time
- Uniform:- No change w.r.t location over specified region
- Example:- Turbine, pump, boilers, condensers, etc.
- Unsteady devices:- Reciprocating engines, reciprocating pumps etc.



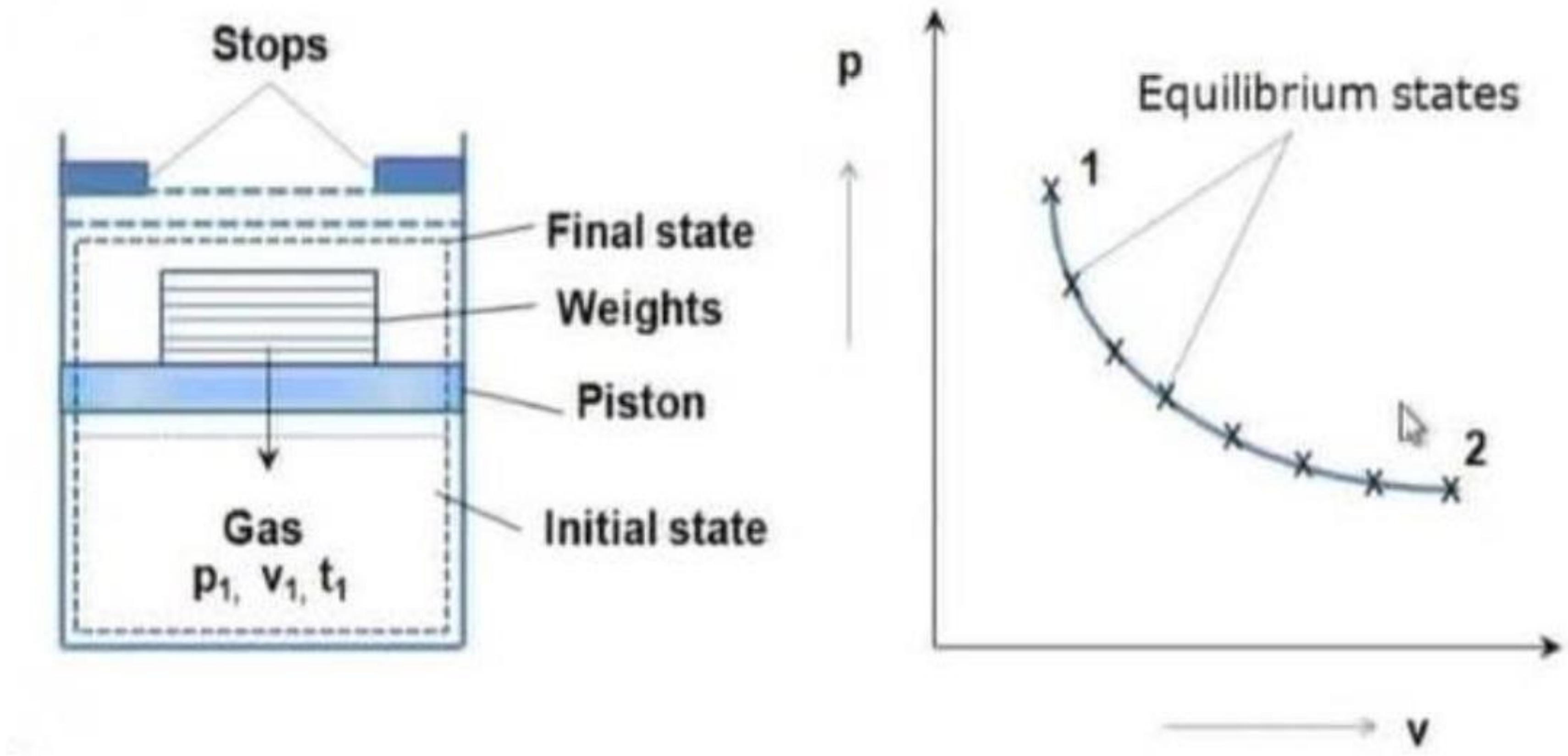
# Unsteady Process

# Thermodynamic Process

| Process | Property held constant |
|---|---|
| Isobaric | Pressure |
| Isothermal | Temperature |
| Isochoric | Volume |
| Isentropic | Entropy |



# Quasi-Static Process



# Quasi-Static Process

- A quasi static process is defined as a process in which the properties of the system depart infinitesimally (extremely small) from the thermodynamic equilibrium state.
- If the properties of the system has finite departures from thermodynamic equilibrium state the process is said to be non-quasi static
- Quasi static process is the succession of thermodynamic equilibrium state while in case of non-quasi static process the end states only represent the thermodynamic equilibrium.

# Quasi-Static Process

- In a process- system remains in equilibrium state at all time
- Every state passed through by the system will be an equilibrium
- Very slow process
- Idealized process
- Also called as Reversible process

# Why Quasi-Static Process?

- Easy to analyse

- Work producing devices deliver most work when they operate in Q-S process

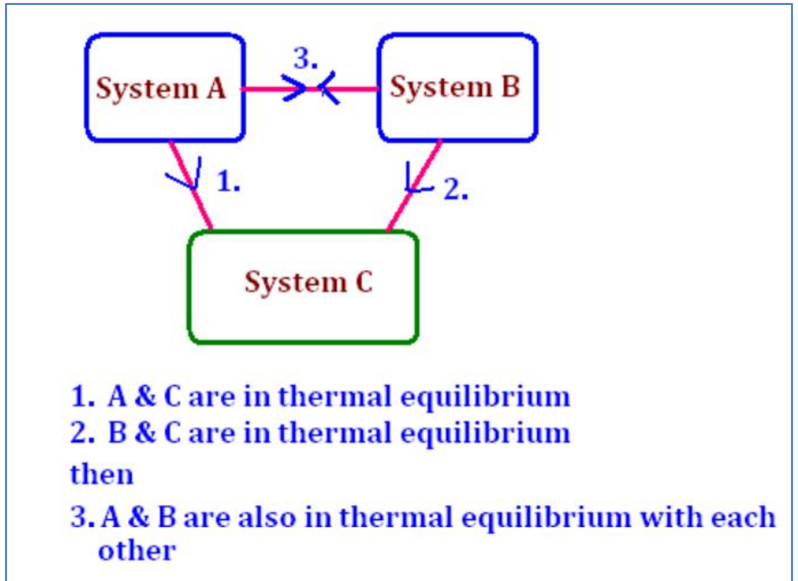
- It acts as standard to which actual process can be compared

# Temperature and Zeroth Law of TD

- Temperature: – is associated with ability to distinguish hot from cold

- Thermal Equilibrium:- Same temperature

- "If two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other"



# Temperature and Zeroth Law of TD

- Serves as basis for validity of temperature measurement
- Temperature scales
- All of them based on Ice point and Steam point
- Ice point:- temperature at which liquid and solid water are in equilibrium under atmospheric pressure
- Steam point:- mixture of liquid water and water vapor (with no air) in equilibrium at 1 atm pressure

# Temperature Scales

- Kelvin $T(\mathrm{K}) = T(°\mathrm{C}) + 273.15$

- Degree Celsius

- Fahrenheit $T(°\mathrm{F}) = 1.8T(°\mathrm{C}) + 32$

# Problem

Calculate temperature values in other scales

- Body Temperature = 98.6$^{o}$F
  - $^{o}$C=
  - K=

- Body Temperature (during fever) = 100.4°F
  - $^{o}$C=
  - K=

# Discussions: Cooling a room using Fridge

# Energy Interactions

- Thermodynamics:- Mainly studies energy interactions and the associated property changes of the system

- In TD:- No information about absolute value of energy, only deals with change in total energy

- Ex: Falling rock example, only height difference not absolute height

# Energy Interactions

- Total Energy = Microscopic + Macroscopic

- **Macroscopic**:- Form of energy system possesses as a whole w.r.t. some outside reference frame

  Ex: KE, PE etc.

- **Microscopic**:- Related to molecular structure of a system and degree of molecular activity, independent of outside reference frames

- Sum of all microscopic forms of energy is called internal energy (U)

# Energy Interactions

Macroscopic Energy

- Kinetic Energy:- Energy that a system possesses as a result of its motion relative to some reference frame, when all parts of a system move with same velocity, KE is expressed as,

$$\boxed{\mathrm{KE} = m\frac{V^2}{2} \qquad (\mathrm{kJ})}$$

- Potential Energy:- Energy of a system possesses as a result of its elevation in a gravitational field

$$\boxed{\mathrm{PE} = mgz \qquad (\mathrm{kJ})}$$

# Energy Interactions

- Other forms of Energy
  - Magnetic
  - Electric
  - Surface tension etc.

- Total Energy:

$$E = U + \text{KE} + \text{PE} = U + m\frac{V^2}{2} + mgz \qquad (\text{kJ})$$

$$e = u + \text{ke} + \text{pe} = u + \frac{V^2}{2} + gz \qquad (\text{kJ/kg})$$

- Open systems (Control Volume):- Energy transfer due to fluid flow also exist

# First Law of Thermodynamics

- Conservation of energy principle
- i.e. every bit of energy should be accounted during a process
- Based on experimental observations of Joules
- Basis for studying the relationship among the various forms of energy and energy interactions
- Consequence of First law:- definition of a new property internal energy

# First Law of Thermodynamics

- Change in the energy of a system during a process is simply equal to the net energy transfer to/from the system

- Energy Balance

$$\begin{pmatrix}\text{Total energy}\\ \text{entering the system}\end{pmatrix} - \begin{pmatrix}\text{Total energy}\\ \text{leaving the system}\end{pmatrix} = \begin{pmatrix}\text{Change in the total}\\ \text{energy of the system}\end{pmatrix}$$

or

$$E_{in} - E_{out} = \Delta E_{system}$$

# First Law of Thermodynamics

- Energy change of a system

Energy change = Energy at final state − Energy at initial state

or

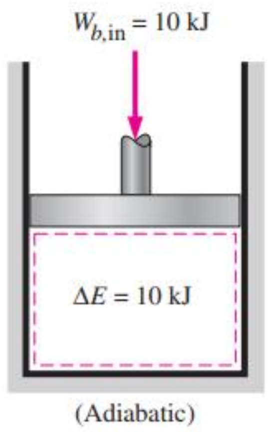
$$\Delta E_{system} = E_{final} - E_{initial} = E_2 - E_1$$

$$\Delta E = \Delta U + \Delta KE + \Delta PE$$

$$\Delta U = m(u_2 - u_1)$$

$$\Delta KE = \frac{1}{2}m(V_2^2 - V_1^2)$$

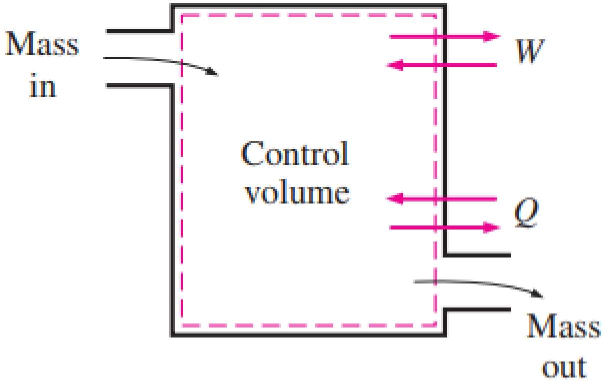
$$\Delta PE = mg(z_2 - z_1)$$




The work (boundary) done on an adiabatic system is equal to the increase in the energy of the system.

# First Law of Thermodynamics

- Mechanism of energy transfer
  - Heat Transfer
  - Work Transfer
  - Mass Transfer




$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) + (E_{mass,in} - E_{mass,out}) = \Delta E_{system}$$

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc., energies}}} \quad \text{(kJ)}$$

or, in the **rate form,** as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\text{Rate of net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{dE_{system}/dt}_{\substack{\text{Rate of change in internal,} \\ \text{kinetic, potential, etc., energies}}} \quad \text{(kW)}$$

# First Law of Thermodynamics:- For a cycle

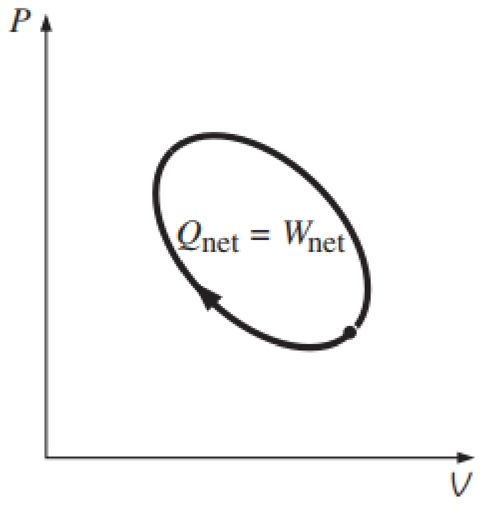
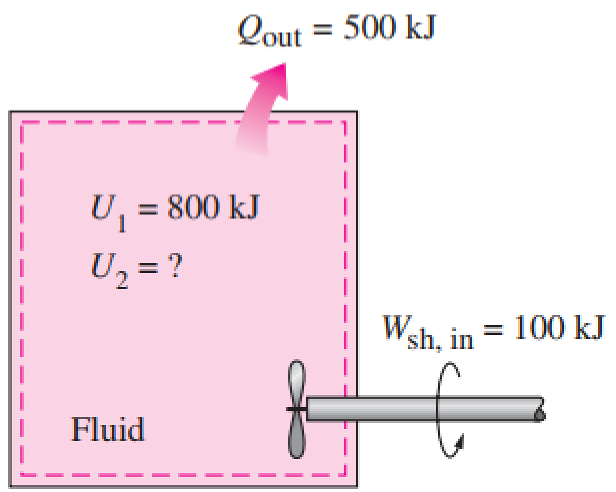



**FIGURE 2–46**

For a cycle $\Delta E = 0$, thus $Q = W$.

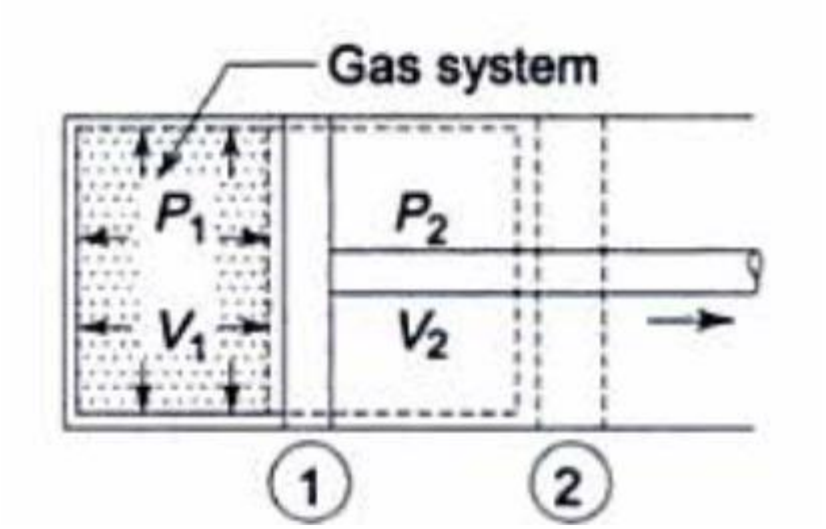
# Problem

A rigid tank contains a hot fluid that is cooled while being stirred by a paddle wheel. Initially, the internal energy of the fluid is 800 kJ. During the cooling process, the fluid loses 500 kJ of heat, and the paddle wheel does 100 kJ of work on the fluid. Determine the final internal energy of the fluid. Neglect the energy stored in the paddle wheel.



# Displacement Work / PdV work

Gas in a cylinder



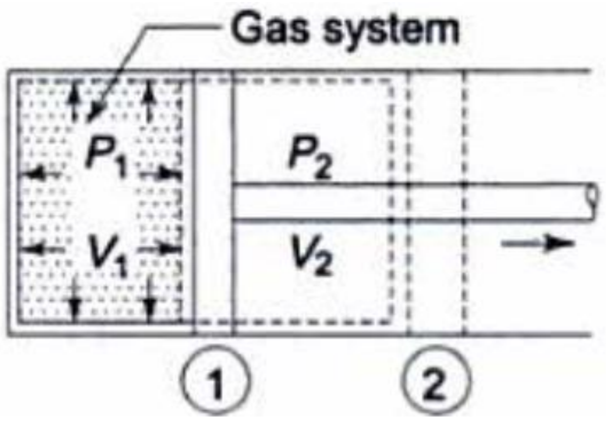

- Initial state: P1, V1
- Final state: P2, V2
- Intermediate states: Assuming Eqm states
- When piston moves at an infinitesimal distance ‘dl’,

$$\text{đ}W = F \cdot \mathrm{d}l = pa\,\mathrm{d}l = p\,\mathrm{d}V$$

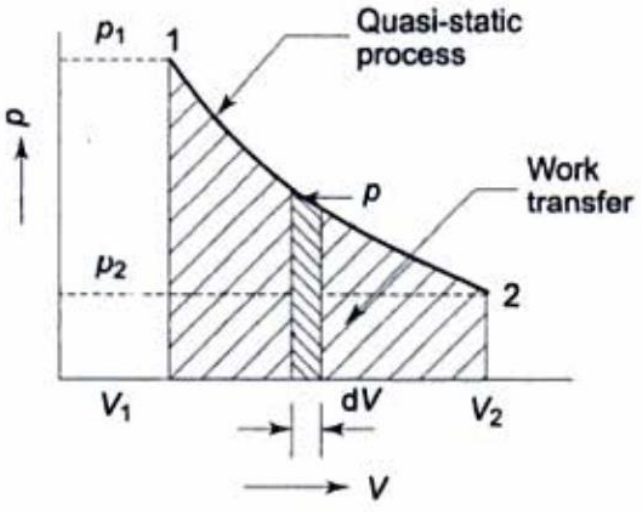
# Displacement Work / PdV work

- Gas in a cylinder

$$W_{1-2} = \int_{V_1}^{V_2} p\,dV$$




- Work done: Area under the p-V diagram
- P and V should be in equilibrium states
- The piston moves infinitely slow so that every state passed through is an equilibrium state
- Integration possible only for Quasi-Static process



# PdV work for various Quasi-Static Processes

a) Constant Pressure Process ($P$ = constant)

b) Constant Volume ($V$ = constant)

c) Isothermal Process ($PV$ = constant)

d) Polytropic Process ($PV^n$ = constant)

# Problem

*An object of 40 kg mass falls freely under the influence of gravity from an elevation of 100 m above the earth's surface. The initial velocity is directed downward with a magnitude of 100 m/s. Ignoring the effect of air resistance, what is the magnitude of the velocity, in m/s, of the object just before it strikes the earth? The acceleration of gravity is* $g = 9.81\ m/s^2$.

*An object of 40 kg mass falls freely under the influence of gravity from an elevation of 100 m above the earth's surface. The initial velocity is directed downward with a magnitude of 100 m/s. Ignoring the effect of air resistance, what is the magnitude of the velocity, in m/s, of the object just before it strikes the earth? The acceleration of gravity is g = 9.81 m/s*$^2$.

**Solution** Since the only force is that due to gravity,

$$\frac{1}{2}mV_2^2 + mgz_2 = \frac{1}{2}mV_1^2 + mgZ_1$$

$$V_2 = \sqrt{2gz_1 + V_1^2}$$

$$= \sqrt{2 \times 9.81 \times 100 + 100^2}$$

$$= 104.4 \text{ m/s} \qquad \textit{Ans.}$$

# Sensible heat Vs Latent Heat

- Sensible Energy:- The portion of internal energy associated with the KE of the molecules

- Average velocity and degree of activity of the molecules are proportional to the temperature of gas

- Latent heat:- energy associated with phase change

# Problem

- Calculate the energy required for heating $0^oC$ ice to $100^oC$ steam. Take latent heat of melting = 335kJ/kg, specific heat of water = 4.2 kJ/kgK, latent heat of vaporization = 2250 kJ/kg.

# First Law of Thermodynamics:- Open system

- Flow work:
  - Closed system: no mass flow across boundary
  - Open system: mass flow exist across boundary

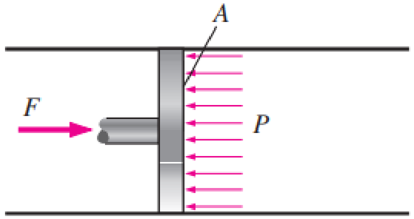



**FIGURE 5–12**

In the absence of acceleration, the force applied on a fluid by a piston is equal to the force applied on the piston by the fluid.

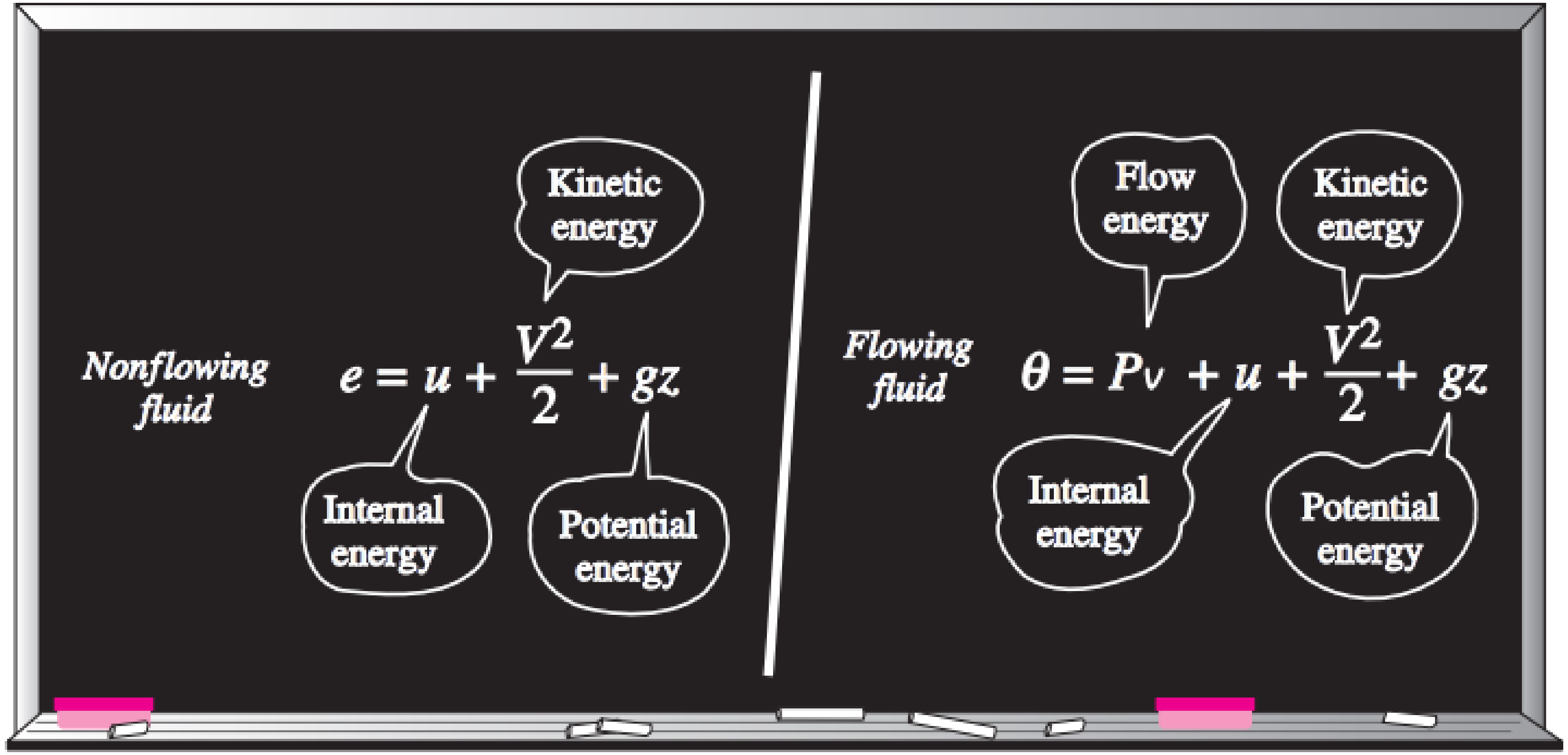
- Flow work: work necessary for maintaining the flow through a control volume

$$W_{flow} = FL = PAL = PV \qquad (kJ)$$

$$w_{flow} = Pv \qquad (kJ/kg)$$

# First Law of Thermodynamics:- Open system

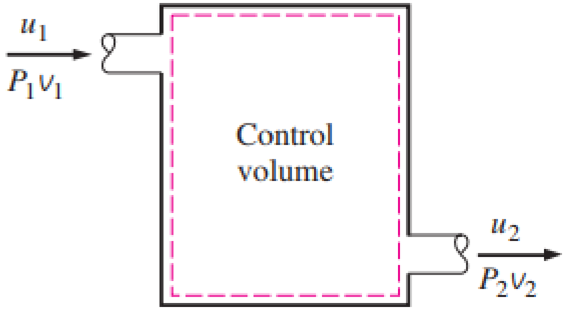
Total Energy of Flowing Fluid



# Enthalpy

- A combined property
- In the analysis of certain types of process, particularly in power generation and refrigeration, we frequently encounter the combined properties ‘u+Pv’
- For the sake of simplicity and convenience, this combination is defined as a new property ‘Enthalpy’

$$h = u + Pv \qquad \text{(kJ/kg)}$$



# Need of 2nd Law of Thermodynamics

# 2nd Law of Thermodynamics

- Process must satisfy 1st law, but satisfying 1st law alone does not ensure that the process will actually takes place

- Eg: Hot coffee in cold room

- Process occurs in certain directions, 1st law places no restrictions on the direction of a process, but satisfying the first law does not ensure that the process can actually occurs

- 2nd Law of TD:- can help us to know whether process can occur or not

- A process cannot occur unless it satisfies both 1st and 2nd Law of TD

# 2$^{nd}$ Law of Thermodynamics

- Process should follow 2nd Law, violation of 2nd Law can easily detected with the help of a property called ‘Entropy’

- 2nd Law also asserts that Energy has quality as well as quantity

- 1st Law:- Quantity of Energy
- 2nd Law:- Quality of Energy

- 2nd Law of TD is also used in determining the theoretical limit for the performance of commonly used engineering systems such as heat engines, refrigerators etc.

# 2nd Law of Thermodynamics

- 1st Law is concerned with quantity of energy and transformation of energy from one form to another with no regard to its quality

- Preserving quality of energy is major concern for engineers, and second law provides necessary means to determine the quality as well as degree of degradation of energy

- In other words, more of high temperature energy can be converted to work, and thus it has higher quality than the same amount of energy at lower temperature

# Thermal Energy Reservoirs

- A hypothetical body with a relatively large thermal energy capacity (mass x specific heat) that can supply/absorb finite amounts of heat without undergoing any change in temperature

- Eg: Ocean, lakes, river, atmospheric air etc.

- Source:- A reservoir that supplies energy in the form of heat

- Sink:- A reservoir that absorbs energy in the form of heat

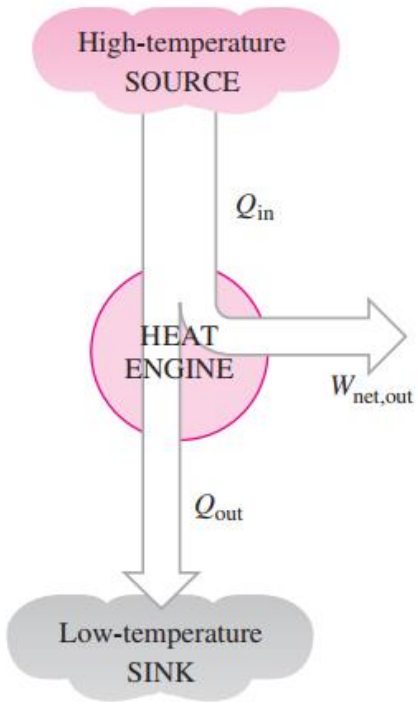
# Heat Engines

- Work can be easily converted to other forms of energy, but converting other forms of energy to work is not that easy

- Heat Engines:- Device which convert heat to work

- Characteristics:
  - They receive heat from higher temperature source
  - They convert part of this heat to work
  - They reject the remaining waste heat to low-temperature sink
  - They operate in a cycle

- Eg: Steam engines



# Thermal Efficiency

- Fraction of heat input that is converted to net work output is a measure of the performance of a heat engine

- How efficiently a heat engine converts the heat that it receives to work

- Petrol Engine = 25 %
- Diesel engine = 40 %
- Gas Turbine = 40 %
- Steam power plant = 60 %

- Even with most efficient heat engines available today, almost one half of the energy supplied ends up in the rivers, lakes, or atmosphere

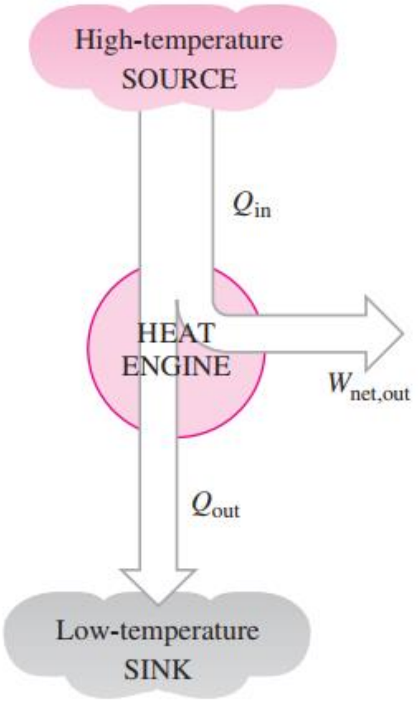
# Thermal Efficiency

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{th} = \frac{W_{net,out}}{Q_{in}}$$

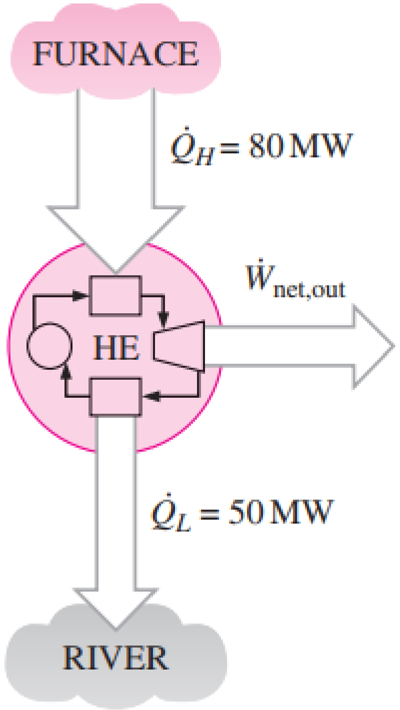
$$\eta_{th} = 1 - \frac{Q_{out}}{Q_{in}}$$

$$W_{net,out} = Q_{in} - Q_{out}.$$



# Problem

Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and the thermal efficiency for this heat engine.



**Solution** The rates of heat transfer to and from a heat engine are given. The net power output and the thermal efficiency are to be determined.

***Assumptions*** Heat losses through the pipes and other components are negligible.

***Analysis*** A schematic of the heat engine is given in Fig. 6–16. The furnace serves as the high-temperature reservoir for this heat engine and the river as the low-temperature reservoir. The given quantities can be expressed as

$$\dot{Q}_H = 80 \text{ MW} \quad \text{and} \quad \dot{Q}_L = 50 \text{ MW}$$

The net power output of this heat engine is

$$\dot{W}_{\text{net,out}} = \dot{Q}_H - \dot{Q}_L = (80 - 50) \text{ MW} = \mathbf{30\ MW}$$

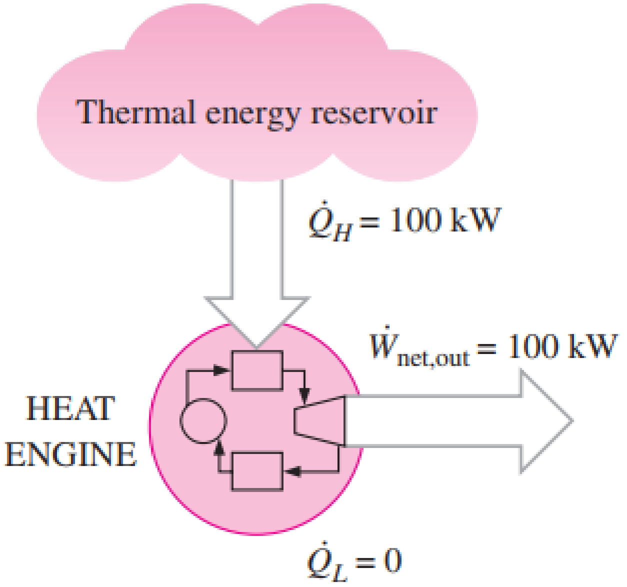
Then the thermal efficiency is easily determined to be

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_H} = \frac{30 \text{ MW}}{80 \text{ MW}} = \mathbf{0.375} \text{ (or } 37.5\%\text{)}$$

***Discussion*** Note that the heat engine converts 37.5 percent of the heat it receives to work.

# $2^{nd}$ Law of Thermodynamics:- Kelvin-Planck Statement

It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.




- No heat engine can have a thermal efficiency of 100 percent, or as for a power plant to operate, the working fluid must exchange heat with the environment as well as the furnace

- Note that the impossibility of having a 100 percent efficient heat engine is not due to friction or other dissipative effects.

# Refrigerator and Heat Pump

- Heat:- high temperature to low temperature
- Transfer of heat from low temperature to high temperature:- device is known as Refrigerator
- Device that transfers heat from a low-temperature medium to a high-temperature one is the heat pump:- Heat pump
- Refrigerators and heat pumps operate on the same cycle but differ in their objectives.

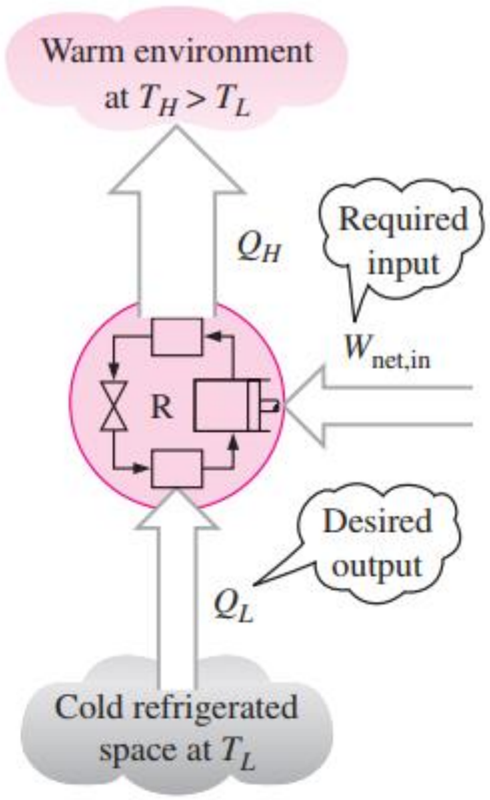
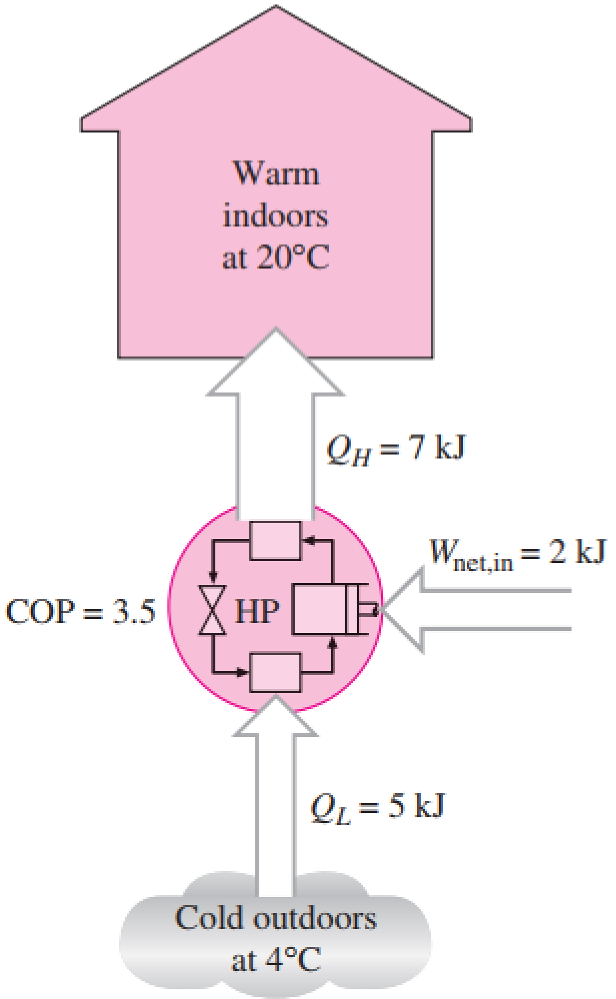



**FIGURE 6–20**

The objective of a refrigerator is to remove $Q_L$ from the cooled space.

# Heat Pump


Warm
indoors
at 20°C
$Q_H = 7$ kJ
$W_{net,in} = 2$ kJ
COP = 3.5
HP
$Q_L = 5$ kJ
Cold outdoors
at 4°C

# Coefficient of Performance

- The objective of a refrigerator is to maintain the refrigerated space at a low temperature by removing heat from it. Discharging this heat to a higher-temperature medium is merely a necessary part of the operation, not the purpose.

- The efficiency of a **refrigerator/heat pump** is expressed in terms of the coefficient of performance (COP)

$$\text{COP}_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_L}{W_{\text{net,in}}} \qquad \text{COP}_{HP} = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_H}{W_{\text{net,in}}}$$

$$\text{COP}_R = \frac{Q_L}{Q_H - Q_L} = \frac{1}{Q_H/Q_L - 1} \qquad \text{COP}_{HP} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - Q_L/Q_H}$$

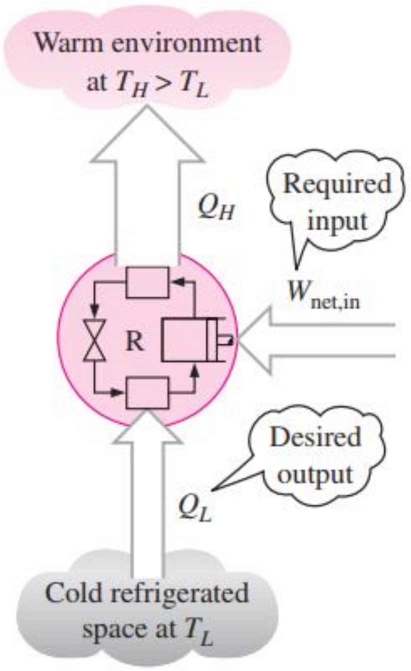



**FIGURE 6–20**

The objective of a refrigerator is to remove $Q_L$ from the cooled space.

# 2nd Law of Thermodynamics:- Clausius Statement

It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

- It simply states that a refrigerator cannot operate unless its compressor is driven by an external power source, such as an electric motor

- Both the Kelvin–Planck and the Clausius statements of the second law are negative statements, and a negative statement cannot be proved

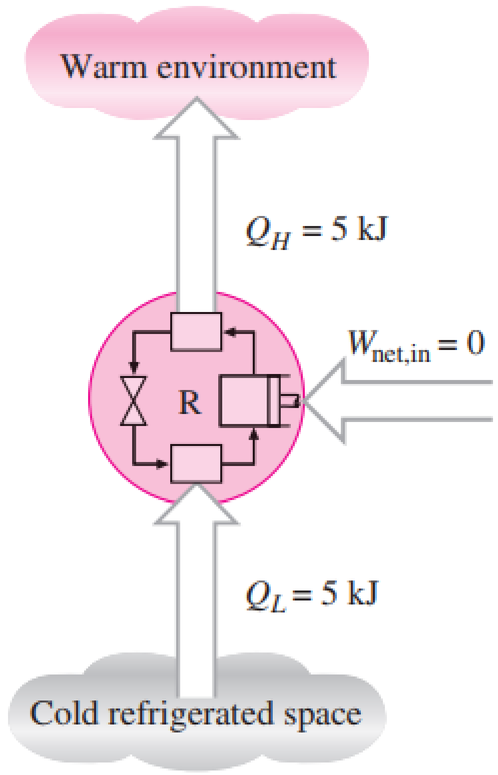


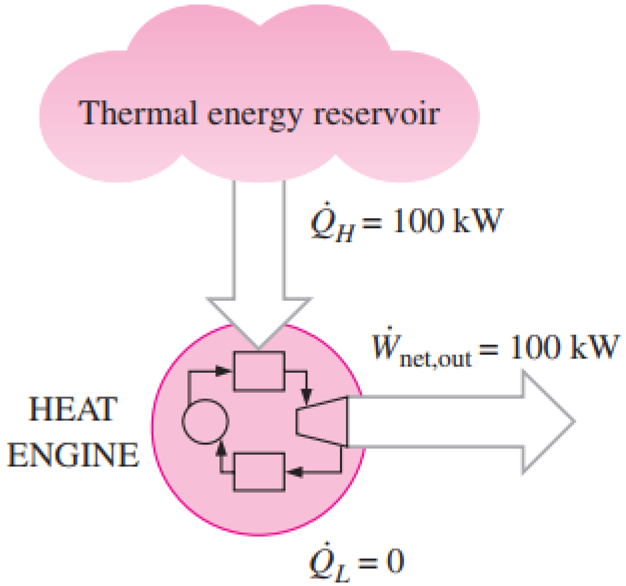

**FIGURE 6–26**

A refrigerator that violates the Clausius statement of the second law.

# 2nd Law of Thermodynamics

## Kelvin-Planck Statement

It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.




## Clausius Statement

It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

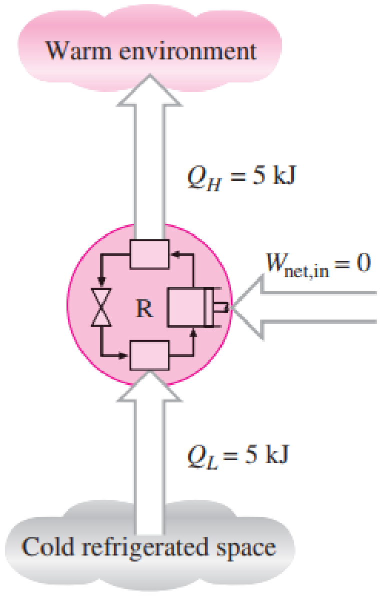
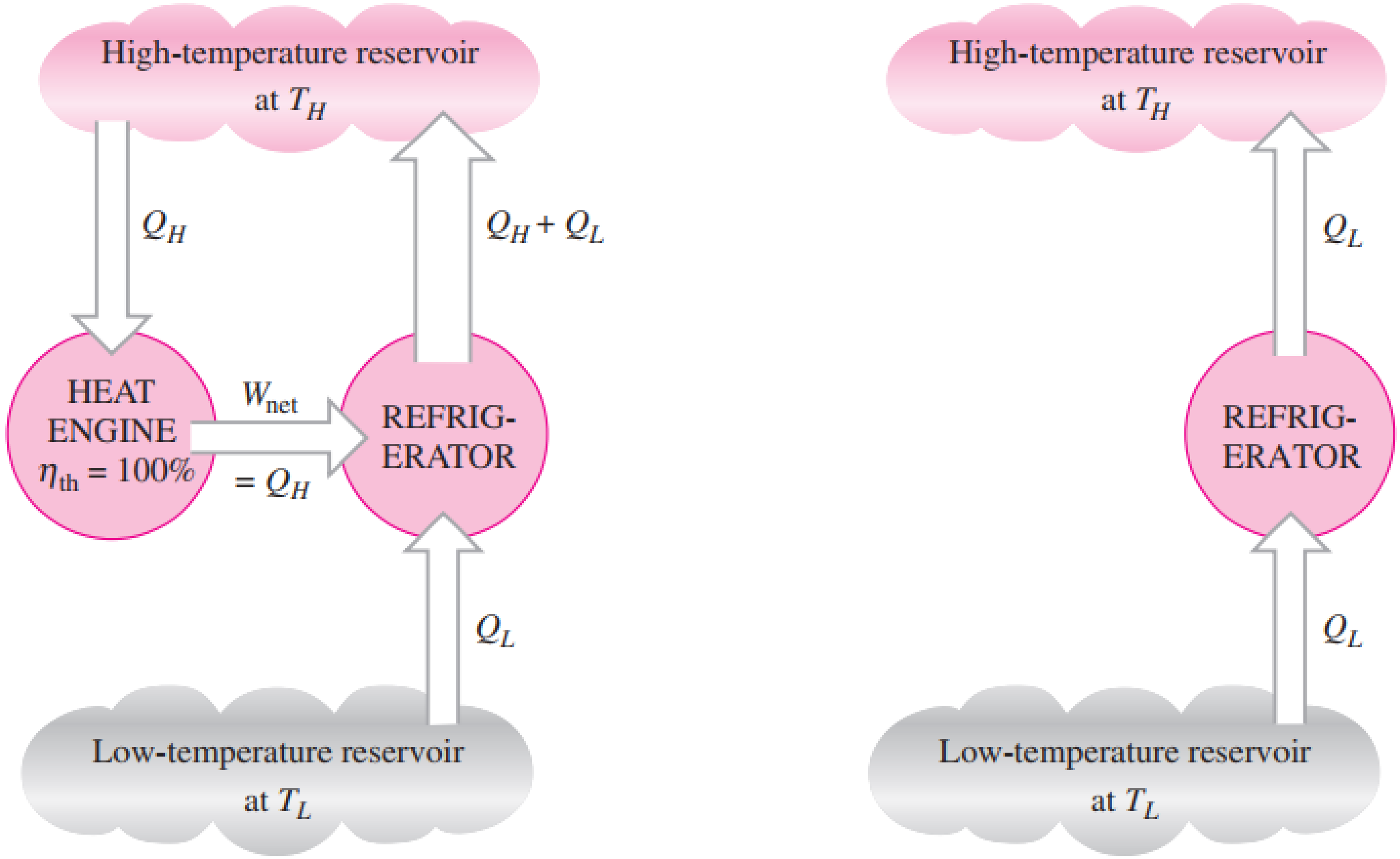



**FIGURE 6–26**

A refrigerator that violates the Clausius statement of the second law.

# Equivalence of the Two Statements




(*a*) A refrigerator that is powered by a 100 percent efficient heat engine

(*b*) The equivalent refrigerator

# PERPETUAL-MOTION MACHINES

- Any device that violates either law is called a perpetual-motion machine,
- A device that violates the first law of thermodynamics (by creating energy) is called a perpetual-motion machine of the first kind (PMM1)
- Fictitious machine that would continuously supply mechanical work without some other forms of energy disappearing simultaneously
- A device that violates the second law of thermodynamics is called a perpetual-motion machine of the second kind (PMM2)

# Carnot Cycle

- Work is done by the working fluid during one part of the cycle and on the working fluid during another part
- The efficiency of a heat-engine cycle greatly depends on how the individual processes that make up the cycle are executed
- The net work, thus the cycle efficiency, can be maximized by using processes that require the least amount of work and deliver the most
- Reversible cycles cannot be achieved in practice because the irreversibilities associated with each process cannot be eliminated
- Best known reversible cycle is the Carnot cycle

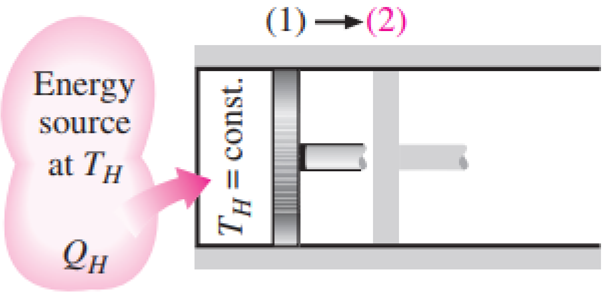
# Carnot Heat Engine

- Theoretical heat engine works on Carnot cycle:- Carnot heat engine

- Composed of four reversible processes
  - Two reversible isothermal processes
  - Two reversible adiabatic processes

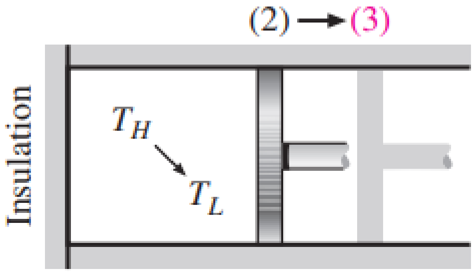
# Carnot Cycle

1. Reversible isothermal expansion



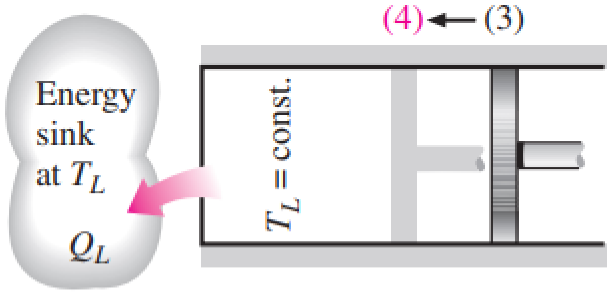

($a$) Process 1-2

2. Reversible adiabatic expansion



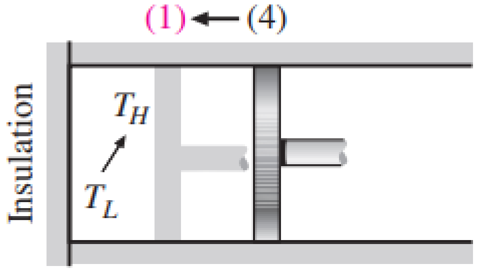

($b$) Process 2-3

3. Reversible isothermal compression



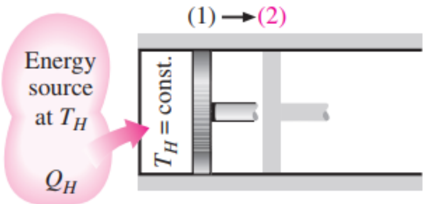

($c$) Process 3-4

4. Reversible adiabatic compression



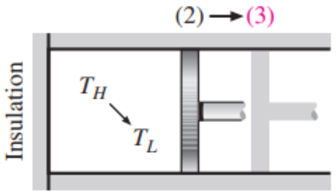

($d$) Process 4-1

# Carnot Cycle



(a) Process 1-2

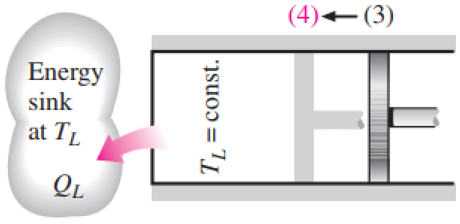
- Reversible Isothermal Expansion (1-2)
  - Gas allowed to expand
  - Doing work on surrounding
  - Isothermal process:- heat is transferred from surrounding to system in order to keep process isothermal



(b) Process 2-3

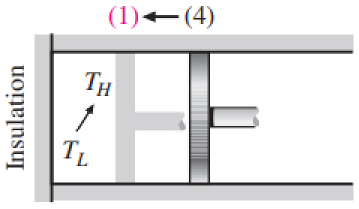
- Reversible Adiabatic Expansion (2-3)
  - The gas continues to expand slowly, doing work on the surroundings until its temperature drops from TH to TL
  - The piston is assumed to be frictionless and the process to be quasi equilibrium, so the process is reversible as well as adiabatic

# Carnot Cycle



(c) Process 3-4

- Reversible Isothermal compression (2-3)
  - System in contact with sink
  - Piston is pushed inward by external force, doing work on gas



(d) Process 4-1

- Reversible Adiabatic Expansion (3-1)
  - Cylinder is adiabatic
  - Gas returns to its initial state (TL -> TH)

# Carnot Cycle

- Area under 1-2-3:- work done during expansion

- Area under 3-4-1:- work done during compression

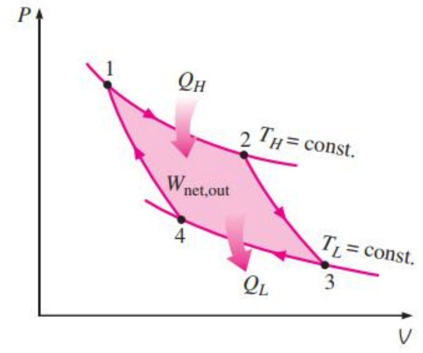
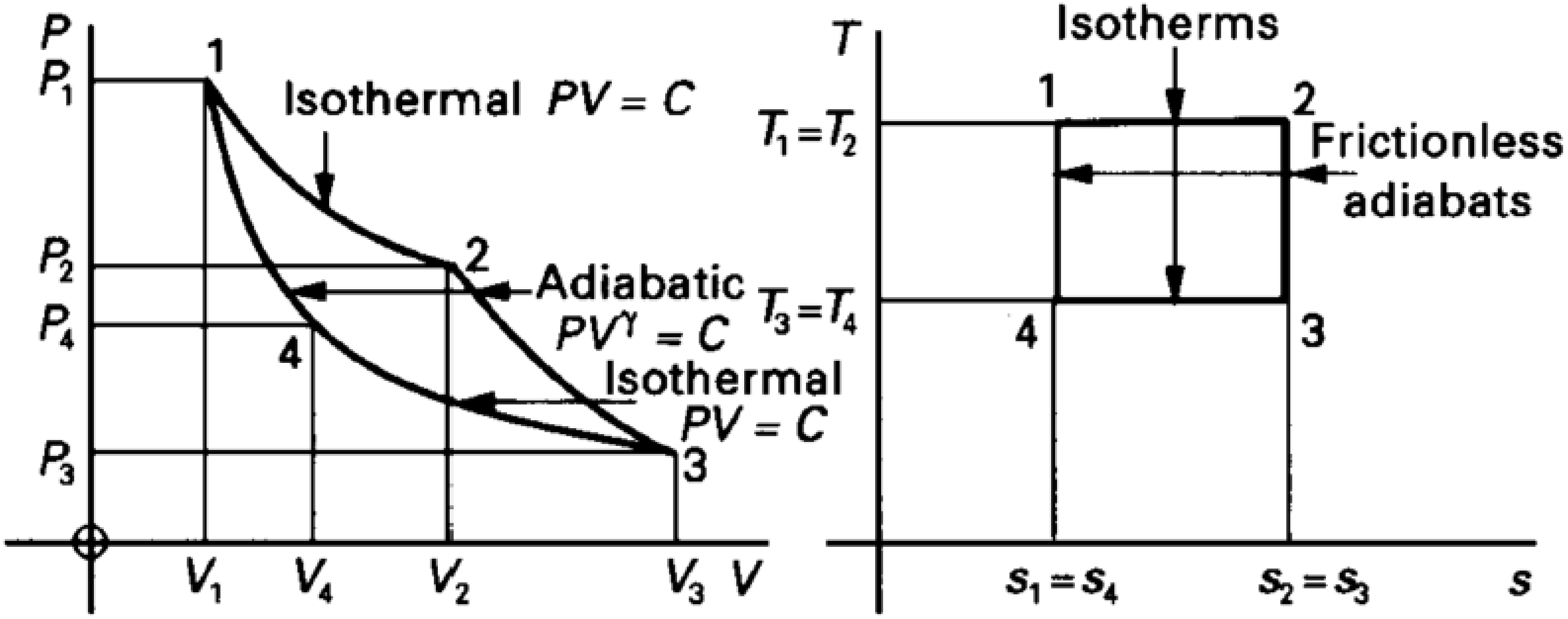



**FIGURE 6–38**
*P-V* diagram of the Carnot cycle.

- Area enclosed by the path of the cycle is the difference between these two:- Net work done during the cycle

- Being a reversible cycle, Carnot cycle is the most efficient cycle operating between two specified temperature limits

# Carnot Cycle



# Efficiency of Carnot Engine

$$\eta_{rev.} = \eta_{max} = 1 - \left(\frac{Q_2}{Q_1}\right)_{rev.} = 1 - \frac{T_2}{T_1}$$

$$\eta_{rev.} = \frac{T_1 - T_2}{T_1}$$

# Why Carnot Cycle is not practical?

- It is impossible to perform a frictionless process
- It is impossible to transfer the heat without temperature potential
- Isothermal process can be achieved only if the piston moves very slowly to allow heat transfer so that the temperature remains constant
- Adiabatic process can be achieved only if the piston moves as fast as possible so that the heat transfer is negligible due to very short time available
- The isothermal and adiabatic processes take place during the same stroke therefore the piston has to move very slowly for part of the stroke and it has to move very fast during remaining stroke
- This variation of motion of the piston during the same stroke is not possible

# Assignment Topics

- Steam generators
- Steam turbines

Thank You